Advanced Programming Practice · 21CSC203P

Unit 3 · Session 8 · SLO 2 — Solved Worksheet

Name: AKARSH MURALI AKARSH MURALI | **Reg. No:** RA2511030010067

Unit – 3 : Session – 8 : SLO – 2 | Close JDBC Resources Properly to Avoid Leaks

I. Fill in the Blanks

1. AutoCloseable — ResultSet, Statement and Connection all implement AutoCloseable, so they can be closed automatically.
2. try-with-resources — from Java 7 onwards resources are closed automatically using try-with-resources.
3. close() — the close() method is used to close a JDBC Connection.
4. resource (connection/memory) leaks — failing to close unused database resources leads to leaks in long-running applications.

II. True or False

1. FALSE — resources must be closed in the reverse order of creation: ResultSet → Statement → Connection.
2. TRUE — closing a Statement automatically closes its currently open ResultSet.

III. Multiple Choice

1. B — the correct manual closing order is ResultSet → Statement → Connection.
2. C — ignoring exceptions during close() is NOT a best practice; close exceptions should be handled or logged.

IV. Programming

1. Find the output: C — rs.close() and stmt.close() succeed, so the program prints "Resources closed."
2. Identify the error: D — Connection.close() and Statement.close() throw the checked SQLException, so the finally block must handle it (try/catch or a throws declaration); additionally the close order should be reversed (stmt before con).